



5.5 Differential equations and Euler's method (HL)

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A differential equation relates a function to its derivative(s), e.g. $dy/dx=f(x,y)$. Some can be solved analytically by separation of variables: $\int 1/g(y) dy = \int f(x) dx$. When an exact solution is difficult or impossible to find, Euler's method numerically approximates the solution given an initial condition (x_0, y_0) and step size h , iterating $y(n+1)=y_n+h \cdot f(x_n, y_n)$ and $x(n+1)=x_n+h$; smaller h generally improves accuracy at the cost of more steps.

Worked example: follow the reasoning

Given $dy/dx=x+y$ with $y(0)=1$, use Euler's method with step size $h=0.5$ to estimate $y(1)$.
Solution: $x_0=0$, $y_0=1$. $f(0,1)=0+1=1$, so $y_1=1+0.5(1)=1.5$ at $x_1=0.5$. $f(0.5,1.5)=0.5+1.5=2$, so $y_2=1.5+0.5(2)=2.5$ at $x_2=1$. Estimate: $y(1) \approx 2.5$.

Quick check · multiple choice

Euler's method's formula for the next y -value, given step size h and $dy/dx=f(x,y)$, is:

- A $y(n+1)=y_n+h$
- B $y(n+1)=y_n+h \cdot f(x_n, y_n)$
- C $y(n+1)=y_n \cdot h + f(x_n, y_n)$
- D $y(n+1)=f(x_n+h, y_n+h)$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A population P satisfies the logistic differential equation $dP/dt=0.1P(1-P/1000)$, [8]
with $P(0)=100$. (a) Use Euler's method with step size $h=1$ to estimate $P(1)$ and $P(2)$. (b) State one advantage of using Euler's method rather than trying to solve this differential equation analytically. (c) State one limitation of Euler's method as an approximation technique.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $f(0,100)=0.1(100)(1-0.1)=9$, so $P(1)\approx 100+9=109$. $f(1,109)=0.1(109)(1-0.109)\approx 9.71$, so $P(2)\approx 109+9.71\approx 118.7$. (b) It gives a usable numerical approximation even when the differential equation (here, non-linear/logistic) is difficult or impossible to solve exactly with standard analytical techniques. (c) It only gives an approximation whose error accumulates over successive steps; accuracy depends heavily on step size h , with smaller h improving accuracy but requiring more computation.

Quick check answer

B. Euler's method takes a step of size h in the direction of the current gradient $f(x_n, y_n)$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.